

Examining the Effects of Tax Service Digitalization, Taxpayer Compliance, and Service Quality on Local Tax Revenue Performance

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ABSTRACT

The digital transformation of tax administration has become a strategic approach to improving taxpayer compliance and optimizing local tax revenue performance. This study aims to examine the influence of tax service digitalization, taxpayer compliance, and service quality on local tax revenue performance at the Regional Revenue Agency (Bapenda) of DKI Jakarta Province. A quantitative explanatory research design was employed using survey data collected from 300 taxpayers representing various local tax categories. Data were analyzed using Structural Equation Modeling-Partial Least Squares (SEM-PLS) with SmartPLS 4. The measurement model evaluation demonstrated satisfactory psychometric properties. All indicators achieved outer loading values above 0.70, confirming indicator reliability. Furthermore, Cronbach's Alpha and Composite Reliability values exceeded 0.70, while Average Variance Extracted (AVE) values were above 0.50, indicating strong reliability and convergent validity. Discriminant validity was also established through the Fornell-Larcker Criterion, Cross Loadings, and Heterotrait-Monotrait Ratio (HTMT), with all values meeting the recommended thresholds. These findings confirm that the proposed measurement model is valid and reliable for assessing tax service digitalization, taxpayer compliance, service quality, and local tax revenue performance. The study provides methodological support for future research on digital tax administration and offers practical implications for tax authorities seeking to enhance service quality, strengthen taxpayer compliance, and improve local tax revenue performance through digital transformation initiatives.

ABSTRAK

Transformasi digital dalam administrasi perpajakan telah menjadi pendekatan strategis untuk meningkatkan kepatuhan wajib pajak dan mengoptimalkan kinerja penerimaan pajak daerah. Penelitian ini bertujuan untuk mengkaji pengaruh digitalisasi layanan perpajakan, kepatuhan wajib pajak, dan kualitas layanan terhadap kinerja penerimaan pajak daerah di Badan Pendapatan Daerah (Bapenda) Provinsi DKI Jakarta. Desain penelitian eksplanatori kuantitatif digunakan dengan memanfaatkan data survei yang dikumpulkan dari 300 wajib pajak yang mewakili berbagai kategori pajak daerah. Data dianalisis menggunakan *Structural Equation Modeling-Partial Least Squares* (SEM-PLS) dengan perangkat lunak SmartPLS 4. Evaluasi model pengukuran menunjukkan sifat psikometrik yang memuaskan. Semua indikator mencapai nilai *outer loading* di atas 0,70, yang menegaskan keandalan indikator. Selain itu, nilai *Cronbach's Alpha* dan Reliabilitas Komposit melebihi 0,70, sedangkan nilai *Average Variance Extracted* (AVE) berada di atas 0,50, yang menunjukkan reliabilitas yang kuat dan validitas konvergen. Validitas diskriminan juga ditetapkan melalui Kriteria Fornell-Larcker, *Cross Loadings*, dan Rasio Heterotrait-Monotrait (HTMT), dengan semua nilai memenuhi ambang batas yang direkomendasikan. Temuan ini menegaskan bahwa model pengukuran yang diusulkan valid dan andal untuk menilai digitalisasi layanan perpajakan, kepatuhan wajib pajak, kualitas layanan, dan kinerja penerimaan pajak daerah. Penelitian ini memberikan dukungan metodologis bagi penelitian selanjutnya mengenai administrasi perpajakan digital serta menawarkan implikasi praktis bagi otoritas perpajakan yang berupaya meningkatkan kualitas layanan, memperkuat kepatuhan wajib pajak, dan meningkatkan kinerja penerimaan pajak daerah melalui inisiatif transformasi digital.

1. Introduction

1.1. Research Background

Tax revenue is the primary source of development financing in many countries, including Indonesia. In the era of rapidly advancing digital transformation, governments around the world are increasingly modernizing their tax administration systems to improve tax collection effectiveness, broaden the tax base, and enhance taxpayer compliance. The Organisation for Economic Co-operation and Development (OECD) emphasizes that the digitalization of tax administration has become a global strategic agenda through the concept of Tax Administration 3.0, which integrates digital technology into all tax administration processes to create a more efficient, transparent, and responsive system [1].

The digital transformation of tax administration has gained greater attention following the increasing demand for improved public services and the optimization of tax revenues in the post-pandemic period. The OECD reports that more than 100 tax administrations worldwide have implemented various digital innovations, including online services, digital identification systems, automated reporting mechanisms, and data-driven technologies to improve compliance and administrative effectiveness [2]. These digital initiatives have been shown to reduce administrative costs, accelerate service delivery, and strengthen taxpayer monitoring and enforcement activities [3].

These developments indicate that successful tax management in the modern era is determined not only by fiscal policy but also by the government's ability to utilize digital technology to improve interactions with taxpayers. According to the Technology Acceptance Model (TAM), users are more likely to adopt technology when they perceive it as useful and easy to use [4]. Therefore, the digitalization of tax services is expected to enhance taxpayer compliance by providing easier access to services, faster administrative processes, and greater transparency within the tax system [5], [6].

In addition to service digitalization, taxpayer compliance is a fundamental factor influencing the success of tax revenue collection. Based on the Theory of Planned Behavior (TPB), taxpayer compliance behavior is affected by individual attitudes, subjective norms, and perceived behavioral control [7]. A high level of compliance contributes to increased tax revenue realization, whereas low compliance can result in the loss of potential regional income that could otherwise be used to finance development programs and public services [8].

Another important factor is the quality of services provided by tax administration agencies. From the perspective of Service Quality Theory, high-quality

services can improve taxpayer satisfaction, trust, and loyalty [9]. Services that are fast, accurate, transparent, and easily accessible encourage taxpayers to fulfill their tax obligations voluntarily. Conversely, inadequate service quality may lead to dissatisfaction and potentially reduce taxpayer compliance levels [10].

In the Indonesian context, local governments play a significant role in optimizing Regional Original Revenue (*Pendapatan Asli Daerah* – PAD) through local tax collection. As the center of national economic activity, the Special Capital Region of Jakarta (DKI Jakarta) contributes substantially to local tax revenues. Therefore, the Regional Revenue Agency (*Badan Pendapatan Daerah* – Bapenda) of DKI Jakarta Province continues to implement various digital innovations to improve service quality and facilitate taxpayers in fulfilling their tax obligations [10]. Data from Bapenda DKI Jakarta Province indicate that regional revenue realization in the first semester of 2024 reached IDR 19.10 trillion, with local tax revenue contributing IDR 16.83 trillion. This achievement highlights the critical role of local taxes in supporting regional development while also demonstrating the substantial revenue potential that can be further optimized through various tax performance improvement strategies [11].

Despite the implementation of numerous digital innovations, challenges in improving local tax revenue performance remain significant. Some taxpayers have not fully utilized the available digital services, compliance levels vary across different types of taxes, and the quality of services perceived by taxpayers may differ. These conditions suggest that the success of digital transformation depends not only on the technology employed but also on taxpayer behavior and the quality of services delivered by local tax authorities [12].

Previous studies have generally examined the relationship between tax service digitalization and taxpayer compliance or the effect of service quality on taxpayer compliance separately [13], [14]. Other studies have primarily focused on the national taxation system rather than local taxation contexts [15]. Furthermore, limited research has simultaneously investigated the effects of tax service digitalization, taxpayer compliance, and service quality on local tax revenue performance within local government institutions, particularly at Bapenda DKI Jakarta Province.

The research gap addressed in this study lies in the integration of three key determinants—tax service digitalization, taxpayer compliance, and service quality—in explaining local tax revenue performance. The novelty of this study also stems from its focus on Bapenda DKI Jakarta Province, a region characterized by a high level of economic activity, a large taxpayer population, and a complex local tax administration

system compared with other regions in Indonesia. Consequently, this research is expected to provide a more comprehensive understanding of the factors influencing local tax revenue performance in the era of digital transformation [16].

Based on the foregoing discussion, this study aims to analyze the effects of tax service digitalization, taxpayer compliance, and service quality on local tax revenue performance at Bapenda DKI Jakarta Province. Theoretically, this study is expected to enrich the literature on digital tax administration, taxpayer compliance, and public service quality. Practically, the findings are expected to provide valuable insights for Bapenda DKI Jakarta Province in formulating strategies to improve local tax revenue through the optimization of digital services, the strengthening of taxpayer compliance, and the enhancement of service quality for the public.

1.2. Literature Review

1.2.1. Tax Service Digitalization and Taxpayer Compliance

Digital transformation in the public sector has become one of the primary strategies adopted by governments to improve the effectiveness of public service delivery [17]. In the context of local tax administration, tax service digitalization refers to the use of information and communication technologies to facilitate taxpayer registration, tax payment, tax reporting, and access to tax-related information [17]. According to the Technology Acceptance Model (TAM), individuals are more likely to adopt a technology when they perceive it as useful (perceived usefulness) and easy to use (perceived ease of use) [18]. When digital tax service systems simplify administrative procedures and reduce transaction costs, taxpayers are likely to perceive greater benefits compared to conventional methods, thereby increasing their level of satisfaction [19].

Furthermore, Expectation Confirmation Theory suggests that user satisfaction emerges when the performance of a service meets or exceeds users' initial expectations [20]. Effective tax service digitalization can therefore enhance taxpayers' positive perceptions of tax administration services [21]. Previous studies have demonstrated that the digitalization of public services contributes significantly to improving user satisfaction [22].

H1: Tax Service Digitalization has a positive and significant effect on Taxpayer Compliance.

1.2.2. Taxpayer Compliance and Service Quality

Taxpayer compliance refers to the extent to which taxpayers fulfill their tax obligations in accordance with applicable regulations [23]. In the context of modern public administration, compliance is influenced not only by monitoring and sanctions but

also by the quality of the relationship between taxpayers and tax authorities [24]. According to the Theory of Planned Behavior (TPB), compliance behavior is influenced by individual attitudes, subjective norms, and perceived behavioral control [25]. Taxpayers who possess a higher level of tax awareness and understanding are more likely to demonstrate compliant behavior [26]. A high level of compliance reflects taxpayers' confidence in a fair and transparent tax system, which may enhance their comfort and satisfaction in fulfilling tax obligations [27]. Empirical studies indicate that higher taxpayer compliance is associated with greater satisfaction with tax systems and tax administration services [28].

H2: Taxpayer Compliance has a positive and significant effect on Service Quality.

1.2.3. Taxpayer Compliance and Tax Revenue Performance.

Service quality refers to an organization's ability to deliver services that meet or exceed customer expectations [29]. In local tax administration, service quality is an important factor that shapes taxpayers' perceptions of government performance [30]. The SERVQUAL model explains that service quality consists of five dimensions: reliability, responsiveness, assurance, empathy, and tangibles [31]. High-quality services can improve taxpayers' positive perceptions of tax institutions and their overall service experience [32]. Professional, efficient, and responsive services can increase taxpayer convenience when accessing tax-related services [33]. In addition, clear and transparent communication can strengthen perceptions of fairness and trust in tax authorities [34]. Numerous studies have identified service quality as a key determinant of taxpayer satisfaction [35].

H3: Taxpayer Compliance has a positive and significant effect on Tax Revenue Performance.

1.2.4. Service Quality and Tax Revenue Performance

Taxpayer satisfaction refers to the overall evaluation of taxpayers' experiences when interacting with tax systems and tax administration services [36]. From a public management perspective, taxpayer satisfaction serves not only as an indicator of service performance but also as a driver of improved tax revenue outcomes [37]. Stakeholder Theory argues that organizational success depends on the ability to meet the needs and expectations of stakeholders [38]. When taxpayers are satisfied with the services they receive, their trust in government institutions tends to increase [39]. Higher taxpayer satisfaction can enhance taxpayer loyalty, voluntary compliance, and willingness to pay taxes accurately and on time [40]. Previous studies have shown that taxpayer satisfaction positively influences taxpayer compliance and contributes to improved tax revenue performance [41].

H4: Service Quality and Tax Revenue Performance.

1.2.5. The Mediating Role of Taxpayer Satisfaction

In this study, taxpayer satisfaction is positioned as a mediating variable that links tax service digitalization, taxpayer compliance, and service quality to local tax revenue performance [42], [43]. The DeLone and McLean Information Systems Success Model explains that system quality, information quality, and service quality influence user satisfaction, which subsequently affects organizational outcomes and benefits [44]. Similarly, Expectation Confirmation Theory suggests that satisfaction is the result of users' evaluations of their service experiences [45]. When taxpayers are satisfied with the services they receive, they are more likely to become loyal users and demonstrate higher levels of voluntary compliance [46]. Several studies have found that satisfaction acts as a mediator in the relationships between digitalization, service quality, taxpayer compliance, and tax administration performance [47]. This mediating pattern of satisfaction has also been empirically confirmed in other public-service settings, where satisfaction was found to significantly mediate the relationship between organizational input factors and individual performance outcomes [48].

H5: Taxpayer Compliance mediates the effect of Tax Service Digitalization and Service Quality on Tax Revenue Performance.

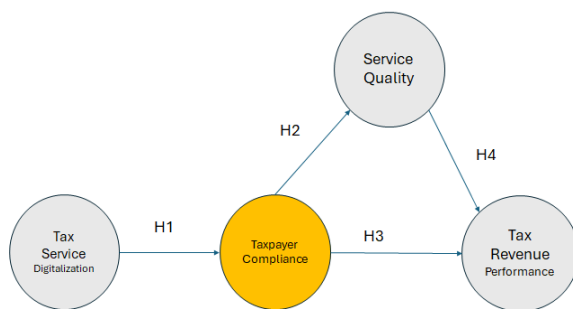


Figure 1. Research Model

2. Research Method

2.1. Research Sample and Study Area

This study employed a quantitative approach using an explanatory research design to examine the causal relationships among Tax Service Digitalization, Taxpayer Compliance, Service Quality, and Local Tax Revenue Performance. The research was conducted at the Regional Revenue Agency (*Badan Pendapatan Daerah* – Bapenda) of DKI Jakarta Province, which is responsible for the administration and optimization of local tax revenues. The study population consisted of all registered taxpayers actively conducting tax-related transactions through Bapenda DKI Jakarta Province. The respondents represented eight categories of local taxpayers, namely Motor Vehicle Tax (PKB) taxpayers, Motor Vehicle Title Transfer Fee (BBNKB)

taxpayers, Hotel Tax taxpayers, Restaurant Tax taxpayers, Entertainment Tax taxpayers, Parking Tax taxpayers, Advertisement Tax taxpayers, and Rural and Urban Land and Building Tax (PBB-P2) taxpayers.

A purposive sampling technique was employed, with respondents selected based on the criterion that they had used Bapenda DKI Jakarta tax services at least once during the previous year. A total of 300 respondents participated in the study. This sample size was considered adequate for Structural Equation Modeling–Partial Least Squares (SEM-PLS) analysis based on the minimum sample size recommendations [49]. Data were collected through both online and offline questionnaire distribution using a five-point Likert scale ranging from 1 (Strongly Disagree) to 5 (Strongly Agree). The selection of Bapenda DKI Jakarta Province as the study site was based on its high level of economic activity, large taxpayer population, and extensive implementation of digital tax transformation initiatives. These characteristics make DKI Jakarta an appropriate setting for examining factors affecting local tax revenue performance in the digital era.

2.2. Variable Measurement

This study employed four main constructs: Tax Service Digitalization (X1), Taxpayer Compliance (X2), Service Quality (X3), and Local Tax Revenue Performance (Y). All variables were measured using reflective indicators developed from relevant theories and previous empirical studies. The Tax Service Digitalization construct was adapted from the Technology Acceptance Model (TAM), emphasizing perceived ease of use, perceived usefulness, service accessibility, process efficiency, data security, and tax information transparency. This construct was measured using six indicators reflecting taxpayers' perceptions of digital tax services implemented by Bapenda DKI Jakarta.

The Taxpayer Compliance construct was developed based on the Theory of Planned Behavior (TPB) and the tax compliance literature. It reflects taxpayers' awareness, adherence to regulations, timeliness of tax payments, accuracy of reporting, understanding of tax regulations, and willingness to comply voluntarily with tax obligations. This variable was measured using six indicators representing taxpayers' compliance behavior. Service Quality was measured based on the SERVQUAL framework, which includes the dimensions of reliability, responsiveness, assurance, empathy, and tangibles [50]. In this study, six indicators were used to assess service quality, including service speed, information accuracy, staff professionalism, service accessibility, staff friendliness, and the ability of staff to resolve taxpayer issues effectively.

Local Tax Revenue Performance was measured based on taxpayers' perceptions regarding the effectiveness of local tax management. The indicators included improvements in tax revenue realization, optimization of tax potential, administrative efficiency, growth in compliant taxpayers, effectiveness of tax supervision, and successful implementation of local tax policies [51]. This construct consisted of six indicators representing the effectiveness of the local taxation system in achieving revenue targets.

All indicators were measured using a five-point Likert scale ranging from 1 to 5, where 1 represented Strongly Disagree, 2 represented Disagree, 3 represented Neutral, 4 represented Agree, and 5 represented Strongly Agree. The Likert scale was selected because it enables comprehensive measurement of respondents' perceptions, attitudes, and experiences and is widely used in behavioral and public administration research. The research instrument was developed based on established theoretical frameworks and previous empirical studies with strong conceptual validity. Prior to the full-scale survey, a pilot test was conducted with a group of respondents to ensure the clarity of language, appropriateness of indicators, and comprehensibility of questionnaire items. All questionnaire statements were designed using simple and easily understandable language suitable for various categories of taxpayers.

In the research model, Tax Service Digitalization was positioned as an exogenous variable influencing Taxpayer Compliance and Local Tax Revenue Performance. Taxpayer Compliance also functioned as a mediating variable explaining the mechanism through which digitalization affects local tax revenue performance. Furthermore, Service Quality was treated as an exogenous variable directly influencing both Taxpayer Compliance and Local Tax Revenue Performance.

The relationships among variables were analyzed using the SEM-PLS approach with SmartPLS 4 software. This method was selected because it is capable of analyzing complex structural models, accommodating latent variables with multiple indicators, and does not require strict assumptions regarding data normality. Moreover, SEM-PLS is particularly suitable for theory development and predictive research.

The conceptual model consisted of direct relationships between Tax Service Digitalization and Taxpayer Compliance, Tax Service Digitalization and Local Tax Revenue Performance, Service Quality and Taxpayer Compliance, Service Quality and Local Tax Revenue Performance, and Taxpayer Compliance and Local Tax Revenue Performance. In addition, indirect effects were examined through the mediating role of Taxpayer Compliance. The assessment of latent constructs involved evaluating both the measurement model (outer model) and the structural model (inner model).

The outer model assessment ensured that indicators appropriately represented their respective latent constructs, whereas the inner model assessment evaluated the strength of relationships among variables and the predictive capability of the proposed model.

The collected data were analyzed using a bootstrapping procedure with 5,000 subsamples to obtain t-statistics and p-values for hypothesis testing. Hypotheses were evaluated at a significance level of 5%, with acceptance criteria of t-statistics greater than 1.96 and p-values less than 0.05. The results of these analyses were used to address the research objectives and test all proposed hypotheses.

2.3. Validity and Reliability Assessment

Instrument validity was assessed through convergent validity and discriminant validity using the SEM-PLS approach in SmartPLS 4. Convergent validity was evaluated based on outer loading values and Average Variance Extracted (AVE). Indicators are considered valid when their outer loading values exceed 0.70, although values between 0.60 and 0.70 may still be acceptable during the early stages of model development. Furthermore, each construct should achieve an AVE value greater than 0.50, indicating that more than 50% of the variance in the indicators is explained by the underlying latent construct [52].

Discriminant validity was assessed using the Fornell–Larcker Criterion and the Heterotrait–Monotrait Ratio (HTMT). A construct is considered to demonstrate adequate discriminant validity when the square root of its AVE exceeds its correlations with other constructs and when the HTMT value is below the recommended threshold of 0.90. These validity assessments ensure that each indicator accurately measures its intended construct and does not overlap substantially with other constructs in the research model.

The reliability of the measurement instrument was evaluated using Cronbach's Alpha, Composite Reliability (CR), and rho_A values. A construct is considered reliable when Cronbach's Alpha and Composite Reliability values exceed 0.70, indicating satisfactory internal consistency. Composite Reliability was used as the primary reliability measure because it is generally considered more appropriate than Cronbach's Alpha in SEM-PLS analysis. In addition, rho_A values above 0.70 were used as supporting evidence of construct reliability. After the measurement model satisfied all validity and reliability requirements, the structural model was evaluated through the examination of the coefficient of determination (R^2), predictive relevance (Q^2), effect size (f^2), and the significance of path coefficients using the bootstrapping procedure. Through these procedures, the study aims to ensure a high level of accuracy and reliability in explaining the factors

influencing local tax revenue performance in the era of digital transformation.

3. Result and Discussion

This section presents the empirical findings of the study, commencing with a rigorous evaluation of the measurement model (outer model) specifically for the

lower-order constructs. The primary objective of this initial evaluation is to establish the reliability and validity of the measurement instruments prior to assessing the structural relationships. The detailed configuration of these construct measurements and their respective indicators is visually depicted in Figure 2 as the initial model structure.

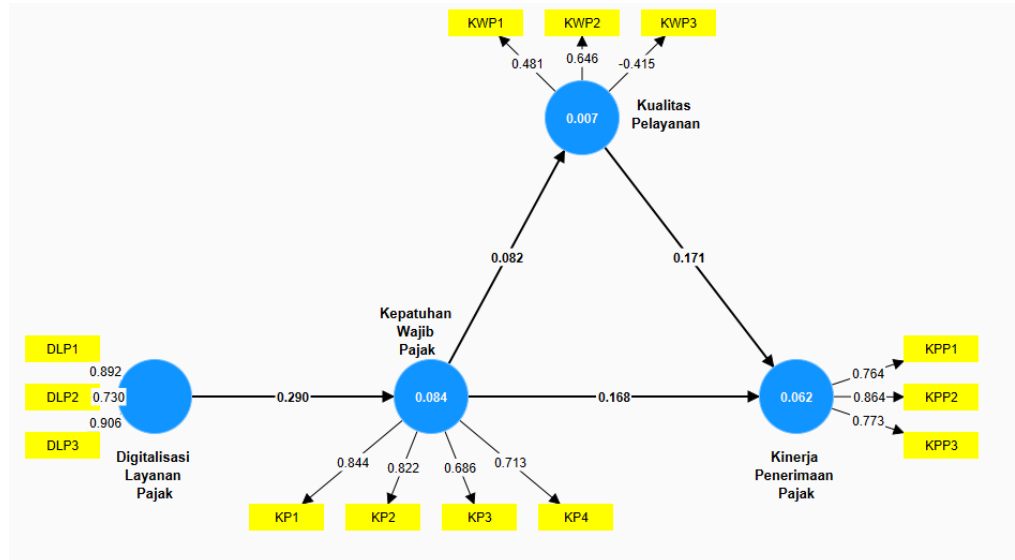


Figure 2. Model Structure

3.1. Indicator Reliability (Outer Loadings)

The first step in evaluating the measurement model (outer model) involves examining the outer loadings of each indicator. Outer loading values indicate the extent to which an indicator reflects its underlying latent construct. Higher outer loading values demonstrate a stronger relationship between the indicator and the construct it is intended to measure, indicating a higher degree of shared variance. Indicators with outer loading values of 0.80 or higher are considered to exhibit strong indicator reliability [53]. High outer loading values suggest that the indicators adequately represent the corresponding construct and contribute significantly to the measurement model. The results of the outer loading analysis are presented in the following Table 1.

Table 1. Outer Loading

Outer Loading	DLP	KP	KPP	KWP
DLP1	0.814			
DLP2	0.802			
DLP3	0.838			
KP1		0.821		
KP2		0.845		
KP3		0.859		
KP4		0.845		
KPP1			0.740	
KPP2			0.834	
KPP3			0.783	
KWP1				0.834
KWP2				0.783
KWP3				0.806

Based on the convergent validity test results presented in Table 1, it can be observed that all indicators have outer loading values greater than or equal to 0.70. Indicators with outer loading values above 0.70 are considered to have adequate indicator reliability and are capable of representing their respective latent constructs effectively. Therefore, all indicators included in this study satisfy the convergent validity requirements and can be retained for further analysis [54].

3.2. Cronbach's Alpha and Composite Reliability (Internal Consistency Reliability)

The next step in evaluating the measurement model (outer model) is the assessment of internal consistency reliability. This assessment is conducted using Cronbach's Alpha and Composite Reliability values. Cronbach's Alpha measures the degree of consistency among indicators within a construct, while Composite Reliability takes into account the different outer loading values of the indicators and is therefore considered a more robust reliability measure in SEM-PLS analysis. Cronbach's Alpha and Composite Reliability values should exceed 0.70 to demonstrate satisfactory internal consistency. Values above this threshold indicate that the indicators consistently measure the same underlying construct [55].

The results presented in Table 2. indicate that all latent constructs satisfy the reliability requirements. This conclusion is supported by the fact that the Cronbach's

Alpha values range from 0.753 to 0.794, while the Composite Reliability values range from 0.858 to 0.880, all of which exceed the recommended threshold

of 0.70. Therefore, all constructs included in this study can be considered reliable and demonstrate adequate internal consistency.

Table 2. Cronbach's Alpha and Composite Reliability (Internal Consistency Reliability)

Construct reliability and validity	Cronbach's alpha	Composite reliability (rho a)	Composite reliability (rho c)	AVE
DLP	0.753	0.756	0.858	0.669
KP	0.794	0.795	0.880	0.709
KPP	0.781	0.803	0.872	0.629
KWP	0.755	0.805	0.870	0.626

3.3. Convergent Validity (Average Variance Extracted – AVE)

Convergent validity refers to the extent to which a construct is able to explain and represent its associated indicators. In SEM-PLS analysis, convergent validity is commonly assessed using the Average Variance Extracted (AVE). An AVE value greater than 0.50 indicates that a construct explains more than 50% of the variance of its indicators, thereby demonstrating adequate convergent validity [56].

Based on the results on Table 2, the AVE values for the constructs range from 0.626 to 0.850, indicating a strong ability to explain the variance of their respective indicators. In addition, the AVE values for variables X1, X2, and X3 range from 0.782 to 0.809, which also reflects satisfactory convergent validity. Overall, all AVE values exceed the recommended threshold of 0.50, indicating that the constructs possess adequate to strong convergent validity and are capable of representing their indicators effectively.

3.4. Discriminant Validity

Discriminant validity assesses the extent to which a construct is empirically distinct from other constructs in the research model. This evaluation ensures that each construct captures a unique phenomenon and does not overlap excessively with other constructs. Common approaches used to assess discriminant validity include the Fornell–Larcker Criterion, Cross Loadings, and the Heterotrait–Monotrait Ratio (HTMT) [57]. The first assessment of discriminant validity is based on the Fornell–Larcker Criterion. According to this criterion, the square root of the AVE for each construct should be greater than its correlations with other constructs. Meeting this requirement indicates that a construct shares more variance with its own indicators than with other constructs in the model.

Table 3. Fornell–Larcker Criterion

Fornell-Larcker criterion	DLP	KP	KPP	KWP
DLP	0.818			
KP	0.653	0.842		
KPP	0.842		0.747	0.735
KWP	0.708			0.791

Based on the results presented in the Table 3, the square root of the Average Variance Extracted (AVE) for each construct is greater than its correlations with the other constructs. This finding indicates that all

constructs satisfy the Fornell–Larcker Criterion and demonstrate adequate discriminant validity. The next criterion to be evaluated is the cross-loading analysis. According to this criterion, the outer loading of each indicator on its associated construct should be higher than its cross loadings on other constructs. This requirement ensures that each indicator is more strongly related to its intended construct than to any other construct in the research model. The loading factor values are presented in the following Table 4.

Table 4. Cross Loadings

Cross loadings	DLP	KP	KPP	KWP
DLP1	0.814	0.478	0.541	0.715
DLP2	0.802	0.562	0.571	0.833
DLP3	0.838	0.556	0.621	0.858
KP1	0.545	0.821	0.617	0.675
KP2	0.579	0.845	0.615	0.654
KP3	0.526	0.859	0.656	0.633
KP4	0.702	0.745	0.657	0.724
KPP1	0.725	0.721	0.721	0.678
KPP2	0.743	0.756	0.712	0.759
KPP3	0.597	0.469	0.740	0.609
KWP1	0.594	0.607	0.834	0.679
KWP2	0.491	0.595	0.783	0.602
KWP3	0.561	0.681	0.806	0.668

Based on the results shown in Table 4, the outer loading value of each indicator is higher than its corresponding cross loading values on other constructs. This finding indicates that each indicator is more strongly associated with its intended construct than with any other construct in the model. Therefore, the cross-loading criterion for discriminant validity is satisfied. Another important criterion for assessing discriminant validity is the Heterotrait–Monotrait Ratio (HTMT). HTMT represents the average correlation between indicators across different constructs and is considered one of the most reliable approaches for evaluating discriminant validity. HTMT values should be below 0.90 to demonstrate adequate discriminant validity. Values exceeding 0.90 suggest a lack of distinction between constructs [58].

Table 5. Heterotrait–Monotrait Ratio (HTMT)

HTMT	DLP	KP	KPP	KWP
DLP	0.845			
KP		0.823		
KPP			0.878	
KWP				0.868

Based on the HTMT results presented in Table 5, all HTMT values are below the recommended threshold of 0.90. Therefore, the constructs satisfy the HTMT

criterion and demonstrate adequate discriminant validity. Overall, all constructs satisfy the requirements of the Fornell–Larcker Criterion, Cross Loading analysis, and HTMT assessment. These findings indicate that each construct is empirically distinct from the others and captures a unique concept within the research model. Consequently, all indicators are considered valid and meet the requirements of discriminant validity.

3.5. Implications

The results of the measurement model assessment indicate that all constructs and indicators satisfy the required validity and reliability criteria. Methodologically, these findings suggest that the research instrument is capable of accurately and consistently measuring Tax Service Digitalization (DLP), Tax Compliance (KP), Local Tax Revenue Performance (KPP), and Taxpayer Compliance (KWP). Consequently, the proposed research model demonstrates sound measurement quality, providing a reliable foundation for subsequent structural model analysis.

From a theoretical perspective, the findings support the notion that tax service digitalization is a measurable construct represented by relevant dimensions associated with taxpayer compliance and tax revenue performance. The successful validation of all indicators demonstrates that the selected measurement dimensions effectively capture the underlying theoretical concepts. Therefore, the instrument developed in this study may serve as a useful reference for future research investigating digital transformation in tax administration and its impact on taxpayer behavior.

From a practical perspective, the findings suggest that tax authorities should continue to enhance the quality and utilization of digital tax services to support higher taxpayer compliance and optimize tax revenue collection. Digital systems that are accessible, efficient, and transparent can improve taxpayers' experiences when fulfilling their tax obligations. Furthermore, the results highlight the importance of continuously improving digital service quality through technological innovation, enhanced system security, and taxpayer education programs to maximize the benefits of digitalization and strengthen tax revenue performance.

3.6. Recommendation

Based on the findings that all constructs and indicators satisfy the required validity and reliability standards, this study recommends that tax authorities continue strengthening the implementation of digital tax services by developing systems that are more user-friendly, responsive, and secure. Improvements in digital service quality should be accompanied by adequate technological infrastructure and effective support services, such as help desks, to ensure that taxpayers

can utilize digital tax services efficiently. In addition, continuous taxpayer education and awareness programs should be implemented to improve understanding and encourage greater participation in digital tax administration.

For policymakers, the findings provide valuable insights for designing strategies aimed at improving taxpayer compliance through digital approaches. Policies that promote easier access to tax services, simplified administrative procedures, and greater integration of technology-based services are expected to enhance taxpayer compliance and contribute to the optimization of tax revenue collection. Regular evaluations of digital service effectiveness should also be conducted to ensure that tax systems remain responsive to taxpayers' needs and technological developments.

For future researchers, it is recommended to expand the research model by incorporating additional variables that may influence taxpayer compliance and tax revenue performance, such as digital literacy, information system quality, trust in tax authorities, tax awareness, or service quality. Future studies may also broaden the geographical scope and increase the sample size to improve the generalizability of the findings. Furthermore, the use of longitudinal designs or mixed-method approaches may provide a more comprehensive understanding of the long-term effects of tax service digitalization on taxpayer behavior and tax revenue performance.

4. Conclusion

Based on the indicator reliability assessment, all indicators measuring Tax Service Digitalization (DLP), Tax Compliance (KP), Local Tax Revenue Performance (KPP), and Taxpayer Compliance (KWP) achieved outer loading values above 0.70. These results indicate that each indicator adequately reflects its corresponding construct and demonstrates satisfactory convergent validity. Therefore, all indicators included in this study are considered valid and can be retained in the measurement model without requiring elimination. Furthermore, the internal consistency reliability assessment showed that all constructs achieved Cronbach's Alpha and Composite Reliability values exceeding the recommended threshold of 0.70. These findings indicate that the research instrument possesses strong internal consistency and is capable of producing stable and reliable measurements. The Average Variance Extracted (AVE) values for all constructs were also above 0.50, demonstrating that each construct explains more than 50% of the variance of its indicators. Accordingly, all latent variables satisfy the requirements of reliability and convergent validity. The discriminant validity assessment, conducted using the Fornell–Larcker Criterion, Cross Loading analysis, and the Heterotrait–Monotrait Ratio (HTMT), also produced satisfactory results. The square roots of the

AVE values were greater than the correlations between constructs, the outer loading of each indicator exceeded its cross loadings on other constructs, and all HTMT values remained below the maximum threshold of 0.90. These findings confirm that each construct possesses unique empirical characteristics and measures a distinct concept within the research model. In conclusion, the measurement model (outer model) satisfies all required validity and reliability criteria. Therefore, the model is suitable for proceeding to the evaluation of the structural model (inner model) and the testing of the proposed hypotheses.

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